

The Greenhouse Effect

Pupil sheet

Name: _____

Date: _____

1. What I see on the screen

The screen shows three cards: **CO₂** (ppm), **Light** (lx) and **Temperature** (°C). **Temperature has no chart** — I record it every minute from the card.

Be careful what the experiment proves. A closed jar under a lamp warms up *mainly because it has a lid*: the lid stops the warm air leaving. To say anything about CO₂ I need **a control** — a second, identical jar, treated identically, with no extra gas.

Scatter of the rig (empty jars, 10 min) = _____ °C A difference smaller than that *cannot be told apart*.

2. What I do on my own

First I predict — **about the *cause*, not the result**. The jar will warm up mainly because of:

☐ the lid (it stops convection) ☐ the walls (they absorb light) ☐ the CO₂ (it absorbs infrared)

because _____

I check the symmetry before starting. Light: jar A _____ lx, jar B _____ lx. Do they differ by under 5%? ☐ yes ☐ no

Experiment 1 — the lid. Both jars with ordinary air; one with a lid, one without.

Minute	With lid (°C)	No lid (°C)	Difference
0			
10			
20			
30			

Verdict (circle one): detected / borderline / not detected

Experiment 2 — CO₂. Both with lids. **I make the same gesture in both** — if I put a cup in one, I put one in the other.

CO₂: the gas jar _____ ppm

the control _____ ppm

Minute	With CO ₂ (°C)	Control (°C)	Difference
0			
10			
20			
30			

Verdict (circle one): detected / borderline / not detected

3. The conclusion I write

I predicted that, and the data show that _____

A garden greenhouse and the atmosphere warm differently, because _____

The very fact that the sensor can measure CO₂ proves that _____

Exit ticket

What warmed the jar, and how do you know?